

2-zero

$$\begin{cases} X_{3j} = X_{1j} - X_{14} & j = 5, \dots, n-1 \\ X_{3n} = -X_{14} \\ X_{2j} = X_{3j} + X_{2n} & j = 5, \dots, n \end{cases}$$

1 BP

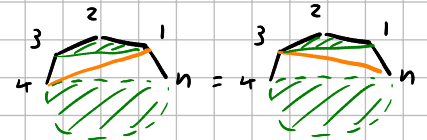
← already set $X_{3n} = -X_{14}$

$$S = \frac{1}{X_{14}} (a_{14} - a_{3n}) + \sum_{j=5}^{n-1} \frac{a_{1j}}{X_{1j}} + \sum_{j=5}^{n-1} \frac{a_{3j}}{X_{3j}} + \sum_{j=4}^n \frac{a_{2j}}{X_{2j}} = 0$$

$$0 = \lim_{X_{14} \rightarrow 0} X_{14} S = a_{14} - a_{3n}$$

so

$$\boxed{a_{14} = a_{3n}} \\ (a_{14} = a_{3n})$$



so $S = \sum_{j=5}^{n-1} \frac{a_{1j}}{X_{1j}} + \sum_{j=5}^{n-1} \frac{a_{3j}}{X_{3j}} + \sum_{j=4}^n \frac{a_{2j}}{X_{2j}} = 0 \rightarrow$ all zeros independently.

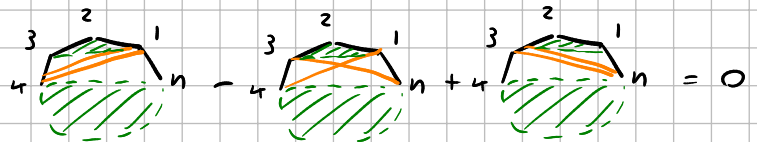
2 BP

$$\frac{1}{X_{14}^2} (a_{1414} + a_{3n3n} - a_{143n})$$

$$+ \frac{1}{X_{14}} \left[\sum_{j=5}^{n-1} \sum_{i=1}^3 \frac{1}{X_{ij}} (a_{14ij} - a_{3nij}) + \frac{a_{1424}}{X_{24}} + \frac{a_{142n}}{X_{2n}} - \frac{a_{3n24}}{X_{24}} + \frac{a_{3n2n}}{X_{2n}} \right]$$

+ - - -

$$\rightarrow a_{1414} + a_{3n3n} - a_{143n} = 0$$



and $\sum_{j=5}^{n-1} \sum_{i=1}^3 \frac{1}{X_{ij}} (a_{14ij} - a_{3nij}) + \frac{a_{1424}}{X_{24}} + \frac{a_{142n}}{X_{2n}} - \dots = 0$ when $X_{14} = 0$

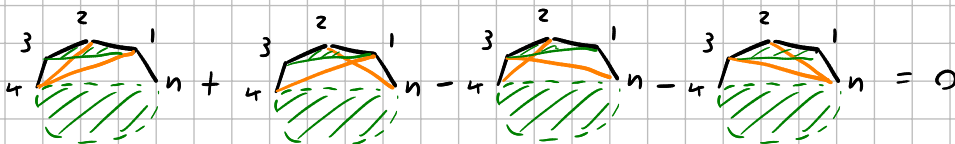
$$X_{3j} = X_{1j}, X_{2j} = X_{1j} + X_{2n} \quad j = 5, \dots, n-1$$

$$\text{and } \boxed{X_{24} = X_{2n}}$$

$$X_{2n} = X_{3n} + X_{24} = X_{24}$$

within here we can set $X_{24} = X_{2n}$, multiply by X_{24} and set to 0

to get $a_{1424} + a_{142n} - a_{3n24} - a_{3n2n} = 0$



then $\sum_{j=5}^{n-1} \sum_{i=1}^3 \frac{1}{X_{ij}} (e_{14ij} - e_{3n ij}) = 0$ when $X_{1j} = X_{2j} = X_{3j}$ (set X_{24} to 0)

$\rightarrow e_{141j} + e_{142j} + e_{143j} - e_{3n1j} - e_{3n2j} - e_{3n3j} = 0$



3 BP

$\frac{1}{X_{14}^3} (e_{141414} - e_{14143n} + e_{143n3n} - e_{3n3n3n})$

$+ \frac{1}{X_{14}^2} \left[\sum_{j=5}^{n-1} \sum_{i=1}^3 \frac{1}{X_{ij}} (e_{1414ij} - e_{143n ij} + e_{3n3n ij}) + \frac{e_{141424}}{X_{24}} + \frac{e_{14142n}}{X_{2n}} + \frac{e_{3n3n24}}{X_{24}} + \frac{e_{3n3n2n}}{X_{2n}} - \frac{e_{143n24}}{X_{24}} - \frac{e_{143n2n}}{X_{2n}} \right]$



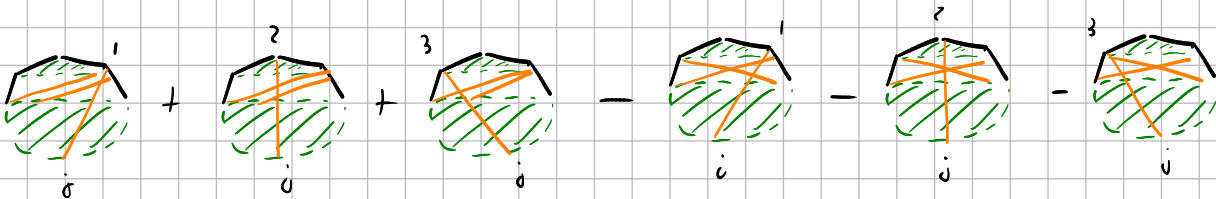
highest power of X_{14}

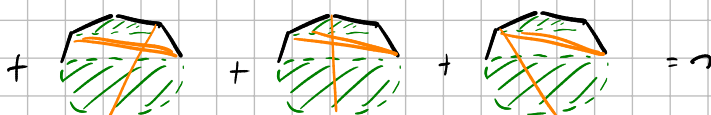
$e_{141424} + e_{14142n} - e_{143n24} - e_{143n2n} + e_{3n3n24} + e_{3n3n2n} = 0$

next power with $X_{24} = X_{2n}$



$\sum_{i=1}^3 (e_{1414ij} - e_{143n ij} + e_{3n3n ij}) = 0, j = 5, \dots, n-1$





3-zero

$$\begin{cases} X_{4j} = X_{15} - X_{15} & j = 6, \dots, n-1 \\ X_{4n} = -X_{15} \\ X_{2\ell} = X_{4\ell} + X_{25} \\ X_{3\ell} = X_{4\ell} + X_{35} \end{cases} \quad \ell = 6, \dots, n$$

1 BP



2 BP

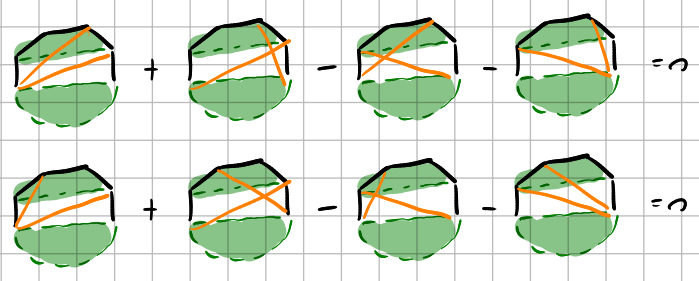


$$\sum_{j=5, n}$$

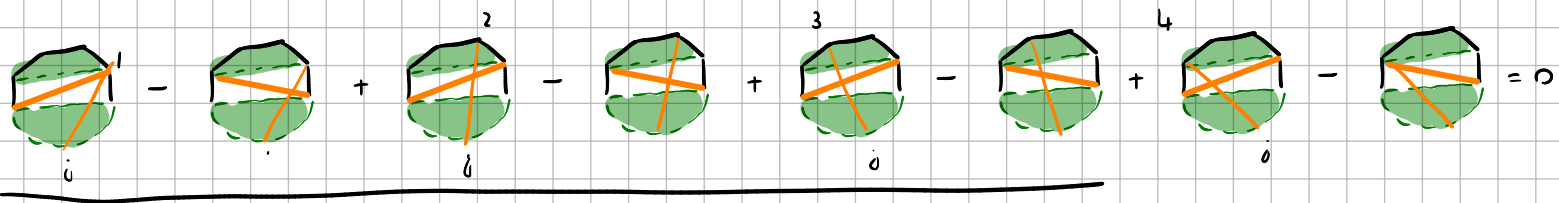
and
$$\sum_{i=1}^4 \sum_{j=6}^{n-1} \frac{1}{X_{ij}} (a_{15ij} - a_{4nij}) + \sum_{i=2}^3 \left(\frac{a_{15i5}}{X_{i5}} + \frac{a_{15in}}{X_{in}} - \frac{a_{4ni5}}{X_{i5}} - \frac{a_{4nin}}{X_{in}} \right) = 0$$

when $X_{15} = 0$ (so $X_{25} = X_{2n}$ and $X_{35} = X_{3n}$)

multiply by $i5$ and in $i \geq 2, 3$: $a_{15i5} + a_{15in} - a_{4ni5} - a_{4nin} = 0 \quad i = 2, 3$



then set $25 = 2n = 35 = 3n = 0$ to get



Reidester nows

$\nearrow \rightarrow \nearrow \nearrow$

